

RF Surveillance: A Case of Homeland and Border Security

Traditional techniques of monitoring individual activities and actions either by manual inspection or by CCTV cameras and the advancement in electronics and radio-frequency-based analysis has opened new avenues for threat detection

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Although security threat can emerge from many sources, the one which is really brutal and catastrophic is terrorism. Terrorism has far reaching consequences like loss of human life and property, and change in behavior pattern of individuals, many of them being irreversible.

Spread of knowledge and education, coupled with strict vigilance, can help contain terrorism and other security threats. However, for robustness in security, electronic threat detection mechanisms are recommended.

The main advantage is the passive monitoring by these electronic solutions, which does not impact routine life of any individual and still ensures safety and security. Generally, terrorist attacks are made in densely-populated or sensitive or tourist areas like bus stations, airports, railway stations, hotels, places of worship and other crowded places.

The frequency spectrum is allotted in India by WPC. This means any frequencies used by anyone who is not legally allotted these raise suspicion, and also any illegal activities carried out in the licensed band are suspect. If the illegal and suspicious frequency and its location are determined by any monitoring system, it can

help to locate and isolate the suspect without much manual intervention. This technique of detecting and locating the illegal and suspicious RF source is called RF surveillance. Here we discuss the RF surveillance technique which can be used both in preventive and tactical ways to effectively counter terrorist threats.

The need

It is not easy to determine illegal activities in communication and mobile services due to their use en-mass. However, communications between terrorists and anti-social elements can be stopped either by shutting down the network or by jamming all the operat-

ing frequencies. This would stop all forms of communications, including security agencies', and is therefore counter-productive. That is why RF surveillance is required and is gaining popularity, as it enhances security perimeter beyond just physical and visual security.

How it works

Radio frequency (RF) surveillance systems must be capable of measuring unknown or unfriendly transmissions and possibly extract the target information content. Direction finding (DF) and geo-location are usually part of RF surveillance signal recovery; also knowledge of the emitter location is offered as part of such systems.

The basic block diagram of a generalised RF surveillance system is as shown in Fig. 1.

The three steps that describe it are as below:

Search. Performs the high-speed signal search operation in the selected frequency band and identifies active signals.

Classify. Performs signal classification operation to sort out the signals of interest based on the selected threshold level, alarm conditions and automatic modulation recognition algorithm.

Monitor. On the signal of interest, performs the demodulation and identifies the intelligence available in it. If required, records the signals and the intelligence

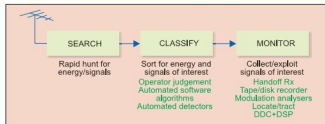


Fig. 1: Basic block diagram of an RF surveillance system



Fig. 2: Multiple RF sensors to determine emitting devices in Hotel Taj Mahal building